Quantum-Assisted Drug Discovery for Lung Cancer

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***Abstract*—The high mortality rate of lung cancer is mainly caused by a late diagnosis and a poor treatment outcome in the late-stage. The process of drug discovery for the lung cancer is very lengthy, as it involves testing of several thousands of chemical compounds to get potential drug candidates. In this work, we introduce a data-driven and quantum-assisted method for computational molecular screening to predict potential lung cancer drugs by evaluating molecular features, identifying latent patterns in compound properties, and correlating molecular descriptors with drug-likeliness. Quantum-assisted computation may provide the opportunity to improve the speed and efficiency of molecular screening compared to classical approaches, especially in the early stages of drug discovery. By combining quantum-assisted methods with classical computational pipeline, the proposed method can rank the most promising compounds and lowers the cost of experiments, and allow us to select the leads in a more rapid and systematic manner. These observations can add to the work in getting right next generation lung cancer drug discovery workflows.**

***Index Terms*—Quantum computing, Drug discovery, Lung cancer, Molecular Docking, Virtual screening, Drug-likeness prediction, Dataset analysis**

I. Introduction

Lung cancer remains a significant global burden and is a leading cause of cancer-related death, with most patients diagnosed at an advanced stage where therapeutic options are of limited efficacy. While significant progress has been made in medical research, current treatments are often unable to confer long-term benefit on patients with progressive disease, indicating a severe unmet need for novel therapeutic modalities.

The process is intricate, laborious and expensive, involving screening of thousands of chemicals, only a minuscule proportion of which turn out to be safe and efficacious. In the early stages of development, important attributes such as effect, safety, stability and drug-likeness are thoroughly tested. Bad decisions at this point can result in the collapse later in development, wasting a great deal of money and time.

Drug discovery has become a vital component of pharmaceutical science allowing the screening and analysis of multiple compounds in an expeditious manner with decreasing burdens on laboratory resources. Nevertheless, traditional computational techniques cannot accurately capture the intricacies of molecular phenomena underpinned by quantum mechanical theories such as electron interactions, protonation, and polarization. Even with pure classical methods, accounting for these effects is nontrivial.

QAC are able to potentially eliminate all these boundaries. Having greater proximity to the actual quantum structure of molecules, such approaches could also improve the accuracy and scale of molecular screening. Quantum-assisted techniques may be integrated with data-driven approaches to allow the researcher to make more informed decisions at the early drug discovery stage. We discuss molecular descriptors in drug discovery for lung cancer and give insight into patterns that show drug-likeness. This coupled strategy can contribute to streamline the lead selection, reduce the costs of the experiments and provide more efficient and reliable lung cancer drug discovery.

II. Literature Review

A. Computational and Quantum Methods in Drug Discovery

Early-stage drug discovery is focused on providing an adequate number of good quality leads, balancing biological activity, drug-likeness and safety, since inappropriate decisions taken early can result in expensive failures [1]. Traditional computational screening methods, particularly ML-based QSAR calculations, have enhanced compound prioritization and experimental burden. More recently, quantum-aided machine learning techniques including quantum kernel-based drug–target interaction models have been shown to provide better performance in early drug screening.

Quantum computing has become a powerful approach for drug design and molecular simulation as it inherently represents molecular wavefunctions and molecular interactions, including electronic interactions that are crucial to the drug binding process [2]. Feasibility of quantum chemical simulations on quantum hardware was established by early studies

whilst hybrid quantum–classical methodologies have facilitated the demonstration of real protein–ligand interactions on NISQ devices [3], [4]. Effective fermion-to-qubit mappings like Jordan–Wigner and Bravyi–Kitaev transformations will continue to be essential to enable the simulation of scalable quantum chemistry [8].

B. Quantum and Structure-Based Approaches in Drug Discovery

Quantum machine learning (QML) is also being investigated alongside to achieve the goal of speeding up drug discovery processes like compound screening and prediction of molecular properties [9]. Hybrid quantum–classical models, with quantum neural networks and generative models are able to achieve better-accuracy predictions and sample chemical space more efficiently, yet limitations in quantum hardware noise and number of available qubits make quantum machine learning a complementary tool for classical workflows [6]. Meanwhile, deep learning-based protein structure prediction has driven SRB in drug design and drug–target interaction analysis [10]. Enhancements in molecular docking including advanced AutoDock Vina and deep learning-based scoring functions have greatly improved the precision and speed of screening [11], while novel quantum optimization methods like NISQ-based QAOA applied to docking problems hold promise as hybrid approaches. In addition, machine-learning methods to predict metal ion binding sites were shown to advance the resolution of protein binding sites [12].

C. Future Directions of Quantum-Assisted Drug Discovery

Quantum computing is promising for protein folding and biomolecular modeling with resource efficient algorithms and feasibility studies in protein structure prediction [13], [14], as well as quantum inspired solutions for better molecular conformation generation in docking studies. Theoretical development provides additional support for the potential of quantum methods in molecular biology and drug design, and successful application on the near-term hardware platform will rely on suitable error mitigation techniques and robust quantum algorithms able to address noise and reliability concerns [15]. The advances on the quantum hardware and programmable molecular simulations will likely further enable the molecular modeling and drug discovery pipelines [7], and the synergy of quantum computing with quantum-inspired optimization and data-driven screening could bring substantial improvements in identifying drug candidates in early stages and reducing time and efforts in lung cancer drug discovery [5].

III. Theoretical Background

A. Drug Discovery

Drug discovery is a process that involves a series of stages and begins at the target identification and validation stage, followed by compound screening to identify molecules that modulate the target. Potential compounds then progress through hit-to-lead and lead optimization phases, where potency, selectivity, and ADMET properties are optimized, and onto preclinical development to assess toxicity and biological activity. It is a time-consuming and costly process and usually takes around 10-14 years before a potential new drug can be marketed after a series of clinical trials (Phases I–IV) to evaluate safety, efficacy, dose, and long-term effects, regulatory approval, and post-marketing surveillance. Early stages of discovery are critical, because they involve the exploration of large chemical spaces and evaluating molecular interactions to identify candidates with adequate efficacy and safety. Modern drug discovery increasingly relies on computational methods for efficiency purposes. Quantum-assisted drug discovery additionally speeds up the early screening process by quickly filtering out potential candidates, potentially reducing the initial discovery timespan from 1–2 years to a couple of weeks, which could in turn allow for swifter progression into downstream preclinical and clinical development while also minimizing cost and resource consumption [32].

B. Quantum Computing

Quantum computing is a computational model based on quantum mechanics wherein information is encoded, manipulated, and stored as quantum bits, called qubits, which can exist as superpositions of states, can be entangled with each other, and can exhibit quantum interference, allowing parallel computations and efficient searches of huge spaces of solutions. Quantum operations can be expressed as reversible unitary operations generated by Pauli-X, Y, Z gates, Hadamard gate for superposition, CNOT gate for entanglement, parametrized rotation gates (Rx, Ry, Rz) for a controlled manipulation of the state. In drug discovery, quantum computing is useful for molecular energy calculation and electronic structure modeling. Hybrid quantum–classical algorithms, including the Variational Quantum Eigensolver (VQE) for ground-state energy determination and the Variational Quantum Classifier (VQC) for quantum machine learning, utilize parameterized quantum circuits with optimizations derived from classical routines. But the limitations of current quantum hardware such as noise, small numbers of qubits and shallow circuit depth severely limit their practical use, so most drug discovery pipelines in the real world make use of classical or quantum-inspired methods.

IV. Methodology

In this section we describe the method used for the discovery of potential EGFR inhibitors through a unified computational pipeline. The entire workflow is illustrated schematically in Figure 1, including data collection, molecular representation, classical and quantum machine learning–based screening, protein and ligand preparation, and molecular docking. The details of each step of the protocol are provided in the following subsections.

A. Dataset and Preprocessing

In this work, we have chosen the epidermal growth factor receptor (EGFR) protein as the target because of its established

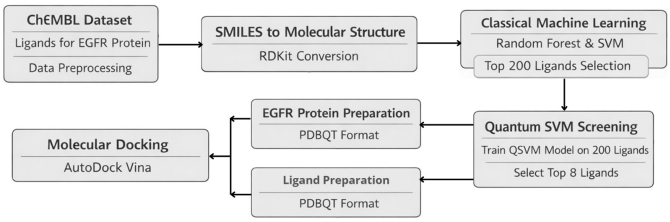

Fig. 1. Proposed Workflow

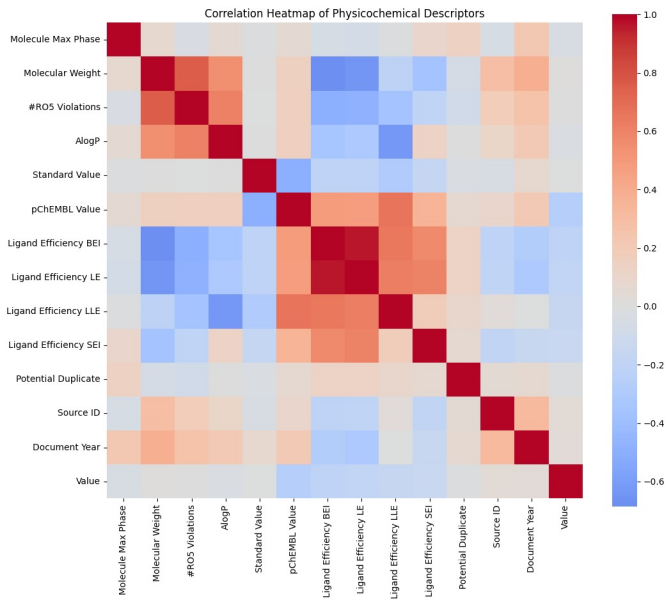

Fig. 2. Correlation Analysis

involvement in lung cancer. A bioactivity dataset of EGFR inhibitors is downloaded from ChEMBL database [33]. The original data set has 25,758 rows (samples) over 48 columns (features) with information about molecules, targets and bioactivities. Then, several processes of data preprocessing were followed, such as dropping the missing values, deleting the duplicated rows and normalizing the numerical features to unify the scales. Following the preprocessing procedure, the dataset was cleaned and distilled to roughly 13,000 entries.

B. Feature Selection

We conducted a correlation analysis on molecular descriptors and features related to bioactivity to reduce dimensionality and tackle multicollinearity. The analysis, Figure 2, showed a high intercorrelation between ligand efficiency–based descriptors suggesting that these descriptors are redundant; as expected, some correlations were also found among physicochemical properties such as molecular weight, lipophilicity, number of RO5 violations.

Feature selection was based on correlation coefficients and descriptors with correlation values above 0.85 was regarded as redundant. Among the strongly correlated ligand efficiency measures, Ligand Efficiency (LE) was kept because of its interpretability and importance in lead optimization. As a result of this procedure, a reduced but informative subset of features was selected, consisting of Molecule ChEMBL ID, SMILES, Standard Type, Standard Units, Target Name, Standard Value, Molecular Weight, ALogP, RO5 violations, and Ligand Efficiency (LE), which enabled computational efficiency without sacrificing significant chemical and activity information.

C. SMILES to Molecular Structure Conversion

In the computer-aided drug design process, the transformation of SMILES strings to molecular structures played an important role in the preprocessing for computational analysis. Compounds without or with ambiguous IC (Standard Value) information were eliminated in the first step of data cleaning.

These remaining SMILES were then processed with the RDKit cheminformatics tool, where the Chem.MolFromSmiles() function creates graph based representations of the molecules with nodes representing atoms, edges between nodes representing bonds and also performs validation of chemical correctness checks (valency, bond types, stereochemistry). These uniform RDKit Mol objects encode key structural elements and facilitate descriptor calculation, fingerprint generation, and 3D conformer generation, and thus serve as the basic building blocks for machine learning, molecular docking and in silico drug discovery methodologies.

D. Classical Machine Learning Screening

After molecular standardization, compound bioactivity was predicted by classical ML models. The molecular descriptors, computed by RDKit, were divided into the training and testing sets in a 80:20 ratio. To reduce the dimensionality of the data and lessen the problem of multicollinearity, PCA was used, which transforms the original correlated features into a smaller number of uncorrelated components while retaining most of the variance.

Two classical methods were considered: Random Forest (RF) and Support Vector Machine (SVM). Random Forest facilitates better prediction by combining several randomized decision trees and SVM determine an optimal separating hyperplane among active and inactive molecules.The SVM model involved a RBF kernel (k), which is represented as:

$$K_{\mathrm{RBF}}(x, x') = \exp\left(-\gamma \|x - x'\|^2\right) \tag{1}$$

where x and x' denote molecular feature vectors, γ controls the kernel width, and $\|x - x'\|^2$ represents the square of Euclidean distance. This kernel allows SVM to capture non-linear relations in the space of molecular descriptors.

The RF and SVM models were reliably capable of classifying bioactivity with accuracies of 79.6% and 78.5%, respectively. Based on the predicted probabilities of being active, the dataset was processed further for quantum machine learning by picking the top 200 molecules out of around 13,000 molecules, leading to a drastic reduction of the computational complexity, while keeping most of the high-potential candidates for further modelling with quantum machine learning and molecular docking.

E. QSVM-Based Ligand Selection

To further screen drug candidates, we applied a Quantum Support Vector Machine (QSVM) on the top 200 ligands selected by classical Random Forest predictions. The encoding of molecular descriptors into quantum states was performed with a ZZ feature map, thus the QSVM could exploit correlations of higher orders. The quantum kernel, which quantifies the similarity of two feature vectors, x and x', is given by:

$$K_Q(x, x') = \left|\langle 0|U^\dagger(x)\, U(x')|0\rangle\right|^2, \tag{2}$$

where $U(x)$ represents the parameterized ZZ feature map:

$$U(x) = \exp\left(i\sum_i x_i Z_i + i\sum_{i<j} x_i x_j Z_i Z_j\right), \tag{3}$$

with Z_i and $Z_i Z_j$ denoting Pauli-Z operators. The first term represents the single-feature information and the second term represents the pairwise interaction, thus the kernel can capture high-order information including complex nonlinear relationships.

The QSVM confidence values and selected key physicochemical properties were used to prioritize the top eight ligands for subsequent molecular docking and binding affinity calculations. Table I list these compounds.

TABLE I
KEY MOLECULAR PROPERTIES OF SELECTED COMPOUNDS

ChEMBL ID	Mol. Wt.	ALogP	PSA	RO5	QED
CHEMBL5900488	551.07	5.52	110.16	2	0.27
CHEMBL5768513	539.03	5.49	100.93	2	0.29
CHEMBL5876257	541.05	5.74	100.93	2	0.27
CHEMBL5990005	553.07	5.44	110.16	2	0.22
CHEMBL5788798	551.07	5.17	110.16	2	0.30
CHEMBL5782179	511.61	4.76	124.72	1	0.33
CHEMBL6060784	582.66	5.74	140.51	2	0.18
CHEMBL4083486	539.64	4.03	104.62	1	0.34

This QSVM-based strategy filters high-confidence candidates to the reduce the computational cost and increase the probability to find strong binders in structure-based drug discovery.

F. EGFR Protein Preparation

The target protein (EGFR) was downloaded from the RCSB Protein Data Bank in PDB format. Figure 3 represents the three-dimensional structure of the EGFR protein in native PDB form, which is the experimental atomic coordinates without any pretreatment. The protein structure was processed using AutoDock Tools prior to docking to accurately reflect the binding site environment and to be suitable for docking simulations.

Protein preprocessing had been done using the protocol of removing crystallographic waters, adding polar hydrogens for modeling hydrogen bonding, and assigning partial charges of kollman for the computation of electrostatic properties. The modified structure was subsequently transformed into PDBQT format (Figure 4), which is the optimized protein for docking.

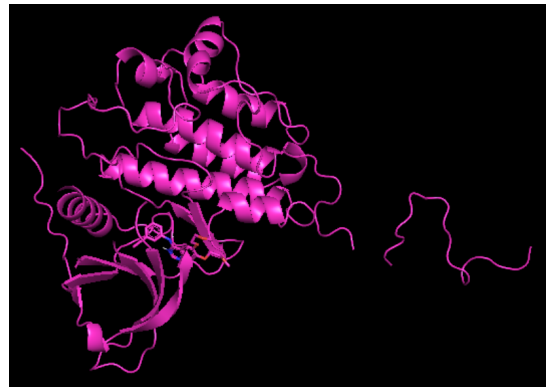

Fig. 3. EGFR protein in PDB format

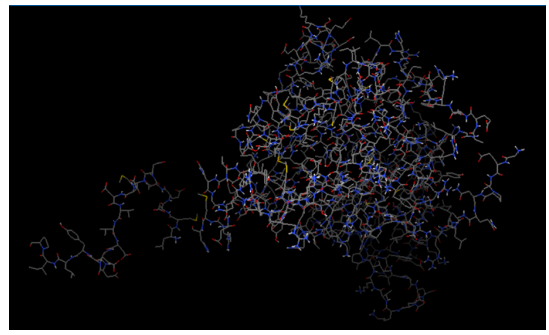

Fig. 4. EGFR protein in PDBQT format after processing

A grid box was placed at the active site of EGFR to restrict the docking search with the center at X = 23.553, Y = 9.725, Z = 59.329 and dimensions of 94 × 102 × 94 points in the X, Y, and Z directions with a spacing of 0.375 Å. Docking calculations were performed with an exhaustiveness of 16, while CPU and verbosity were set to default.

The grid box and protein were generated using a novel standardized procedure, which allows the results of the molecular docking to be reliable, reproducible , and biologically meaningful in subsequent experimental validation.

G. Ligand Preparation

Selected ligands were downloaded from the PubChem database in SDF format which includes the experimentally determined 3D structures used for the accurate spatial description of ligands in docking.

Ligand structures were processed using PYMOL, and converted from SDF to PDB format to be used with the docking software. The ligands were prepared in AutoDock Tools by adding charges, defining rotatable bonds and they were converted to PDBQT format for docking simulation.

These rigorously prepared and energy minimized ligands can now be confidently used to model interactions with the EGFR protein in molecular docking.

H. Molecular Docking

The AutoDock Vina package was used for molecular docking to evaluate the binding energy and stability between

QSVM-selected ligands and EGFR. Docking was performed with the prepared EGFR structure and eight ligand PDBQT files using the same grid box as located before around the active site. Binding energy scores (in kcal/mol) were calculated for each ligand; higher negative values represent stronger binding. All of these ligands from the eight all showed nice affinities, which prove the validity of the previous classical and quantum machine learning–based screening.

Among the docked ligands, CHEMBL5990005 obtains the best docking score which was 9.256 kcal/mol with the lowest docking energy, indicating the highest binding affinity with EGFR. Also other ligands such as CHEMBL6060784, CHEMBL5768513 and CHEMBL5782179 hadhigh binding affinities. These results demonstrate the QC integrated workflow of classical ML, quantum ML and molecular docking for rapid lead identification in drug discovery.

V. Results and Discussions

The results reported here were obtained on the basis of computational screening and molecular docking. However, limitations of the current quantum hardware such as the limited number of qubits and noise-related errors restrict the extent of this work. Such constraints notwithstanding, the results show that the inclusion of quantum-assisted methods in hybrid drug discovery protocols was indeed a viable proposition.

First, classical ML models were used to screen the EGFR inhibitor dataset, with the Random Forest and SVM classifiers attaining 79.6% and 78.5% accuracies, respectively—good enough for the first-level compound filtering. The final 200 ligands were further filtered using QSVM confidence scores, yielding 8 high confidence candidate molecules. This two-stage procedure demonstrates that quantum-enhanced models, when used with classical methods, can provide real value in compound prioritization, despite existing computational and hardware limitations.

TABLE II
QSVM Confidence Scores for Shortlisted Ligands

CHEMBL ID	QSVM Confidence Score
CHEMBL5900488	0.7995
CHEMBL5768513	0.7936
CHEMBL5876257	0.7401
CHEMBL5990005	0.8637
CHEMBL5788798	0.7190
CHEMBL5782179	0.9336
CHEMBL6060784	0.7411
CHEMBL4083486	0.9006

The QSVM confidence scores for the final list of ligands are shown in Table II. Larger values of scores correspond to higher confidence that the compound is active. CHEMBL5990005, which is the top scoring QSVM compound with the lowest docking energy, was considered as the best ligand. Overall, classical screening results were refined using QSVM scores and the compounds with the highest potential were given priority.

The protein EGFR was docked and its binding affinity and stability was evaluated using AutoDock Vina. The docking scores of the eight ligands stated are shown in Table III. CHEMBL5990005 displayed the lowest score (–9.256 kcal/mol), hence most stable complex among ligands, other ligands also obtained good scores less than –8.0 kcal/mol indicating their feasibilities as lead compounds.

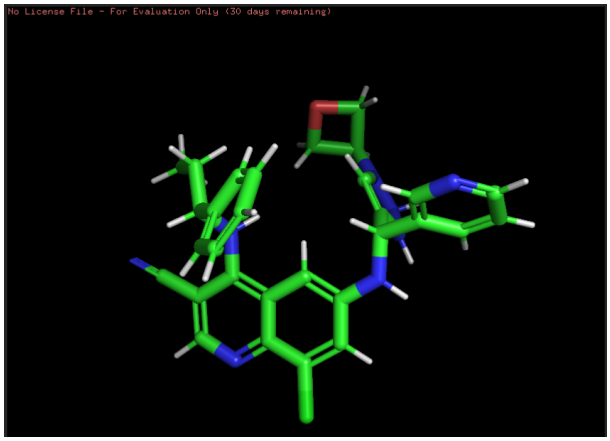

Fig. 5. Docked pose and binding conformation of CHEMBL5990005 within the EGFR active site and binding pocket

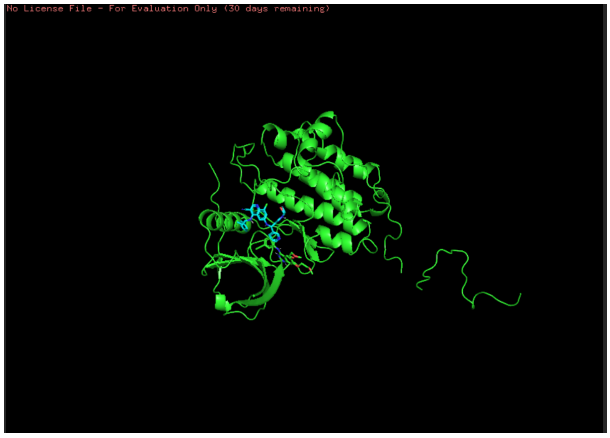

Fig. 6. Docked pose and binding conformation of CHEMBL5990005 within the EGFR active site and binding pocket

TABLE III
Docking Scores of Selected Ligands Against EGFR

CHEMBL ID	Docking Score (kcal/mol)
CHEMBL5900488	-8.195
CHEMBL5768513	-8.734
CHEMBL5876257	-8.060
CHEMBL5990005	**-9.256**
CHEMBL5788798	-8.384
CHEMBL5782179	-8.659
CHEMBL6060784	-8.834
CHEMBL4083486	-8.469

The results of AutoDock Vina for CHEMBL5990005 with several binding modes are tabulated in Table IV. The docking energy for Mode 1 is the minimum with zero RMSD, suggesting a very stable and well converged ligand- protein complex. Fig. 5 depicts CHEMBL5990005 in the binding pocket of EGFR and Fig. 6 shows the stable docked conformation. The

TABLE IV
AUTODOCK VINA DOCKING RESULTS FOR CHEMBL5990005

Mode	Affinity (kcal/mol)	RMSD l.b. (Å)	RMSD u.b. (Å)
1	-9.256	0.000	0.000
2	-9.164	2.446	4.709
3	-9.116	2.749	4.770
4	-9.020	2.916	8.282
5	-8.985	3.327	9.347
6	-8.918	4.036	8.981
7	-8.839	3.680	9.238
8	-8.835	3.221	9.048
9	-8.806	1.992	2.576

ligands uniform direction in the active pocket also highlighted that it was the best molecule in this round of screening.

Typically these outputs reveal that classical screening quantum assisted scoring and the molecular docking are in some cases the most systematic and efficient techniques of early stage drug discovery.

VI. CONCLUSION

In this work, we propose an efficient classical-quantum hybrid computational framework for lung cancer drug discovery involving EGFR. The classical ML techniques, QSVM-based screening, and molecular docking employed in the workflow synergistically reduced the vast chemical space to only a few hit compounds with high potency, enabling to effectively select valid candidates in the early stage. Docking results also validated the high binding affinity of the selected ligands, with CHEMBL5990005 showing maximum stability with EGFR. The results demonstrate that a quantum-assisted approach can improve drug-likeness prediction, reduce development time, and possibly allow for more exact and cost-efficient identification of leads in next-generation drug discovery pipelines.

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